CHO301 CHEMICAL TECHNOLOGY

Making Rocket Fuel on Mars

A Sabatier plant that turns Martian CO_2 and ice into liquid methane and oxygen

$$CO_2 + 4\,H_2 \rightarrow CH_4 + 2\,H_2O$$

625	94.9%	480
kg CH_4 per day	CO_2 conversion	days to 300 t CH_4

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chemical-technology-project.vercel.app

github.com/oyetanishq/chemical-technology-project

Design basis: 350 °C, 5 bar, nickel on alumina

Sabatier reaction

$$CO_2 + 4\,H_2 \rightarrow CH_4 + 2\,H_2O$$

−165 kJ/mol, ΔH: heat must be removed

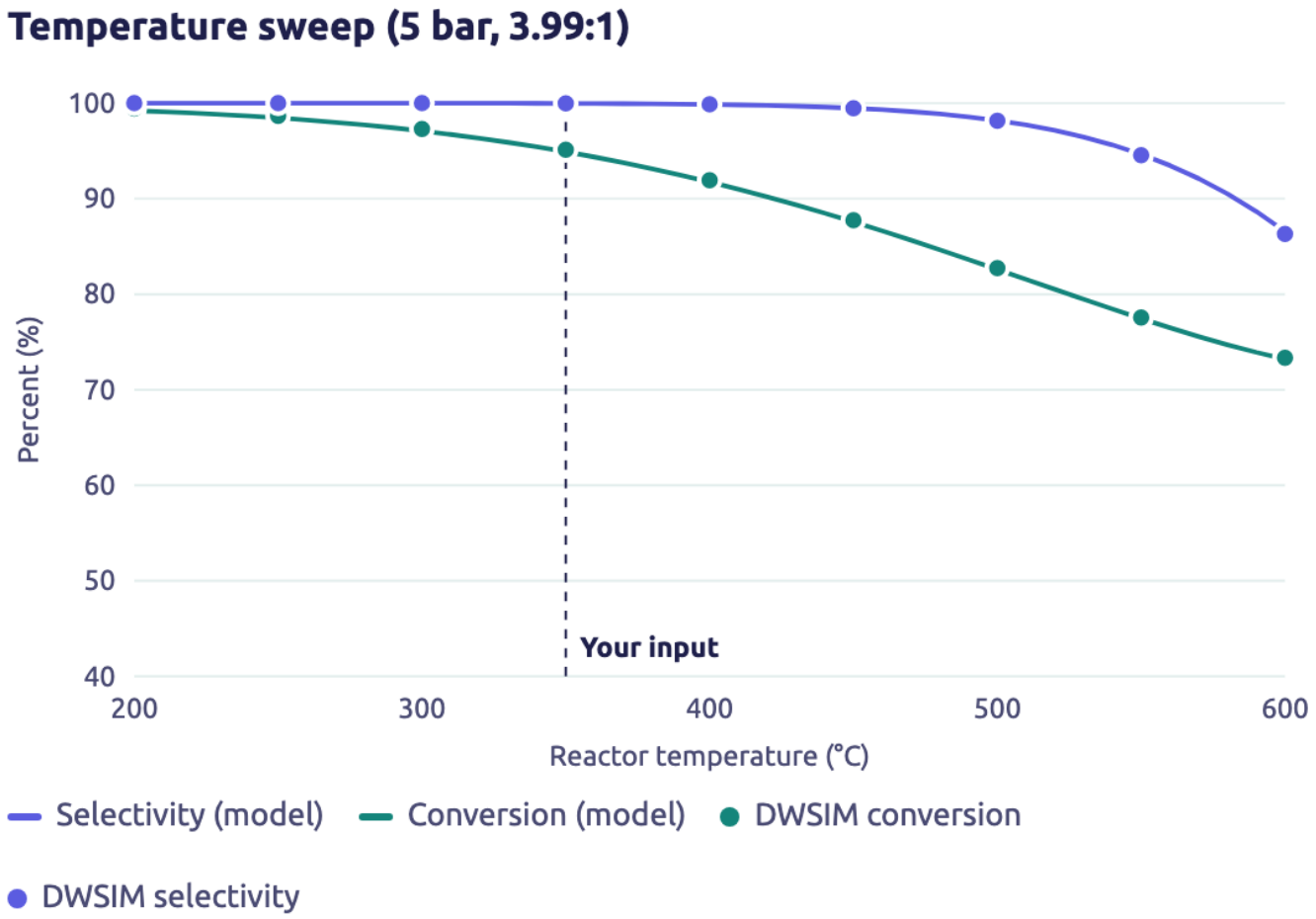
−130.8 kJ/mol, ΔG: strongly favourable

The design tension

Thermodynamics prefers low temperature, since the reaction is exothermic and loses gas volume. Kinetics needs enough heat to get going on nickel. 300–400 °C is the practical compromise.

Parameter	Value	Why
Temperature	350 °C	Middle of the 300–400 °C window. Slow below it, CO forms above it
Pressure	5 bar	Fewer moles of gas on the product side, so pressure helps conversion
Catalyst	Ni on Al_2O_3	Cheap and robust. Ru is more active at low T but costs more
H_2:CO_2 feed	4 : 1	Stoichiometric, so neither feed is wasted. Tuned to 3.99 in the plant
Water by-product	2 mol per CH_4	Returned to the electrolyser, which halves the water demand

Selectivity stays near 100% up to 400 °C



95.1%

conversion, single pass at 4:1

99.98%

CH_4 selectivity at 350 °C

Above 500 °C

Reverse water-gas shift takes over: selectivity is down to 86.3% at 600 °C. Across 300–400 °C it never fell below 99.22%.

Kinetics check

LHHW rate law reaches 92.6% with 1.5 kg catalyst. Its constants still need checking against the paper.

Stoichiometry sets the size of the whole plant

$CO_2 + 2\,H_2O \rightarrow CH_4 + 2\,O_2$ net reaction, once Sabatier water returns to the electrolyser

300 t
CH_4 to make, from 1,350 t of propellant at 3.5:1

823 t
CO_2 from the atmosphere (2.74 kg per kg CH_4)

674 t
net water from ice (2.25 kg per kg CH_4)

1,197 t
O_2 co-produced (3.99 kg per kg CH_4), 147 t surplus

From 39.4 kWh/kg H_2 to the plant's power demand

39.4
kWh/kg H_2
theoretical (HHV)

→

60.6
kWh/kg H_2
at 65% efficiency

→

30.5
kWh/kg CH_4
at 0.503 kg H_2 per kg CH_4

→

793
kW
at 625 kg CH_4/day

Hand calculations agree with two independent models

Metric	Hand calc	Cantera	Month 2 DWSIM	Final DWSIM
CO_2 conversion (%)	100	95.07	95.00 (typed)	95.12
CH_4 selectivity (%)	100	99.97	100	99.98
Sabatier heat (kW)	78.3	81.04	81.13	81.20
Carbon balance (out/in)	1.0000	1.0000	1.00001	1.00000
Hydrogen balance (out/in)	1.0000	1.00002	1.00000	1.00000
Plant mass closure	n/a	n/a	0.00005%	0.00003%

Fuel yield, kg CH_4 per kW per day, depends on what is counted

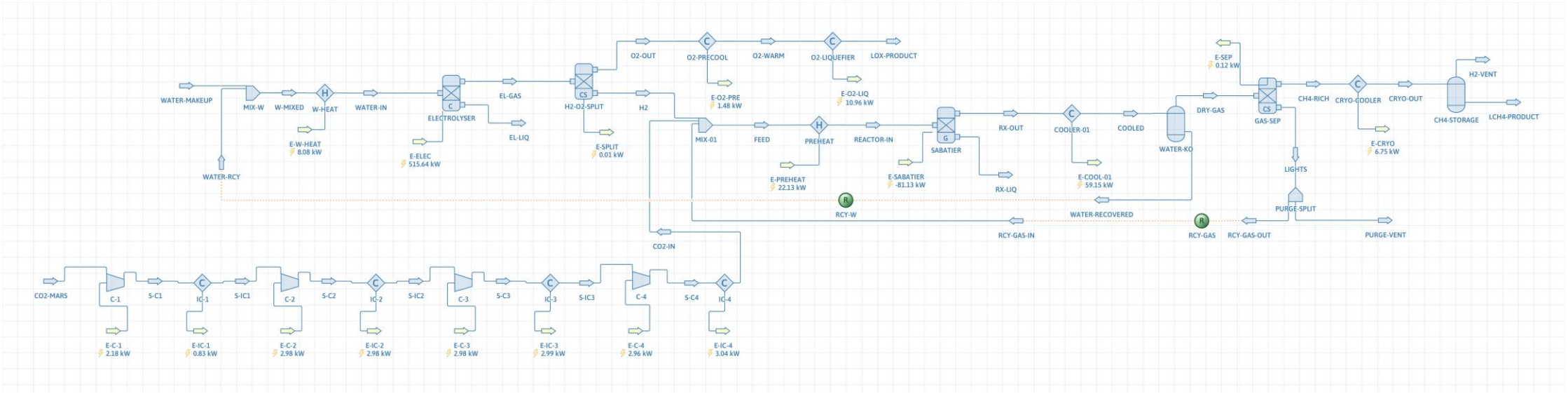
0.79
hand calc: electrolysis only

0.75
final plant: adds compressors and heaters

0.66
estimate: also adds refrigeration at 30% of Carnot

Still open
Choosing 65% or 70% electrolyser efficiency moves yield by about 7%.

The final DWSIM flowsheet connects every unit



−81.1 kW
Sabatier heat removed at full plant scale

11.1 kW
4-stage CO_2 compressor shaft power, 75% efficient

11.0 kW
heat removed in the O_2 liquefier, down to −170 °C

2 recycles
water and unreacted gas, with a 2% purge

Peng-Robinson, pure CO_2 feed, ideal gas separator (99% H_2, 99.95% CO_2), zero pressure drop, no solid CO_2 phase

The mass balance closes and Month 2 gaps are gone

Plant boundary

Stream	T (°C)	P (bar)	kg/h
CO_2 from Mars (in)	−60	0.0065	71.52
Water make-up (in)	25	5	58.68
Liquid CH_4 (out)	−170	5	26.05
Liquid O_2 (out)	−170	5	104.05
Purge vent (out)	25	5	0.10
Total in = total out			**130.20**

0.00003%

plant mass-balance error

0.007 wt%

CO_2 left in the methane, so it will not freeze out

Month 2 to final model

Item	Month 2	Final
Conversion	95%, typed in	94.9%, computed
Throughput	About 610 kg/day	625 kg/day, on target
Gas losses	About 5% vented	Recycled, about 0.1% purged
Water make-up	123.1 kg/h	58.7 kg/h with recycle
Missing units	CO_2 compressor, O_2 liquefier	Both added and solved
O_2:CH_4 ratio	About 4.3, unexplained	4.0, matches stoichiometry

Flight hardware and published studies anchor the design

NASA MOXIE

Perseverance rover, 2021–23

12 g O_2/h

peak rate at ≥98% purity

- 122 g O_2 across 16 runs
- About 100 W, limited by power rather than chemistry
- Proves CO_2 conversion works on Mars

SpaceX Raptor

Starship methalox engine

3.6 : 1

LOX:CH_4 burn ratio by mass

- 350 bar chamber pressure (Raptor 3)
- Isp 327 s at sea level, 350 s in vacuum
- Sets the propellant our plant must deliver

ESA ISRU strategy

Space Resources Strategy, 2019

2030

target for demonstrated ISRU technology

- Runs through 2030
- Sustained Moon presence by 2040
- 23 activities funded in the 2019 call

4 orders

Our plant makes 104 kg O_2/h against MOXIE's 12 g/h, about 8,700 times more. MOXIE shows the principle; this scale-up is the main technology risk.

Our 1,350 t target is a Starship-class plant

Propellant load by architecture

Architecture	Propellant (t)
NASA DRA 5.0 / EMC	30
Mars Direct ERV	96
SpaceX Starship	1,200
This project	**1,350**

	This project	NASA JSC 2024
Plant size	300 t CH_4 over 480 days	30 and 300 t/yr cases
O_2:CH_4	4.0 : 1 natural output	3.5 : 1 fixed target
Electrolysis share	About 95% of electric load	53–62% of plant power
Scope	Reaction to liquid storage	Also ice mining and CO_2 capture

What this means

Our target matches a reusable Starship-class vehicle and is far above small-capsule ascent vehicle studies, so the vehicle class must always be stated.

42.8×

scale-up in CH_4 rate over the NASA reference plant of 14.6 kg/day

26×

scale-up in power over the same plant's 30 kW, the same order as the rate

Three transport scenarios replace a single $1–2B claim

Scenario	Basis	$/kg to Mars surface	Savings on 1,350 t
Optimistic	Published Starship Mars-surface projections	$150–600	About $0.5B
Realistic	Back-solved from the original claim	$700–1,700	About $1.6B
Conservative	Falcon 9 cost to LEO × gear ratio of 3.5–5	$10,500–15,000	About $17B

Transport cost

$$C(Mars) = C(LEO) \times GR$$

GR between 3.5 and 5 covers injection, orbit insertion and landing

Savings per mission

$$S = M(propellant) \times C(Mars)$$

Avoided cost per kg times propellant no longer shipped

Hardware cost (AMCM)

$$C(dev) \propto Q^{0.59} \times M^{0.66}$$

NASA cost model calibrated on 260 aerospace programs

Payback in missions

$$n = CapEx \div (S - Ops)$$

Ops per mission uses the 2.2-year synodic period

The realistic case pays back in about one mission

Payback by transport scenario (1,350 t propellant)

Scenario	Midpoint $/kg	Savings	CapEx $0.5B	CapEx $1.0B	CapEx $1.5B
Optimistic	375	$0.51B	1.26 missions	3.49 missions	8.51 missions
Realistic	1,200	$1.62B	0.33 missions	0.71 missions	1.16 missions
Conservative	12,750	$17.21B	0.03 missions	0.06 missions	0.09 missions

Payback = CapEx ÷ (savings − operations per mission). The project slides use the full cost range; these use midpoints. Propellant counted at the 3.5:1 engine ratio.

$0.5–1.5B

plant CapEx, first-of-a-kind, AMCM-based

$50–150M

operations per year, about 10% of development cost

Stated plainly

If Starship pricing is optimistic, one mission does not cover the plant even at low CapEx: payback is 1.26 missions at best. CapEx must still be rebuilt from the corrected sizing.

Correcting the basis cuts the reactor from 262 L to 25 L

Method

$V = (n R T / P) \div GHSV$

Reactor feed of 8.57 kmol/h, taken at the real 350 °C and 5 bar, with GHSV of 3,600 h^{-1}.

Basis	Volume
Month 2 script: 550 K, 10 bar basis	261.8 L
Actual flow at 350 °C, 5 bar	**24.7 L**
If GHSV is on standard conditions	53.4 L

Reactor volume at GHSV 3,600 h^{-1}

Actual flow at reactor conditions	24.7 L
Standard-condition flow	53.4 L

The answer depends on which basis the GHSV was quoted on. Confirm the basis before quoting one figure.

Before quoting one figure

The answer depends on which basis the GHSV was quoted on. The corrected sizing script is the controlling figure once it is verified.

Reactor cooling needs the largest exchanger surface

Heat-transfer area (m²)

Preheater	2.93
Sabatier reactor	10.79
Water cooler	7.71
Cryo-cooler	0.90
O_2 liquefier	1.46

A = Q ÷ (U × LMTD) with default U = 250 W/m²K and LMTD = 30 K for every unit: placeholders, not a thermal design.

10.8 m²

Sabatier reactor, the largest surface

7.7 m²

water cooler ahead of the knock-out

CO_2 compressor train, 6.5 mbar to 5 bar

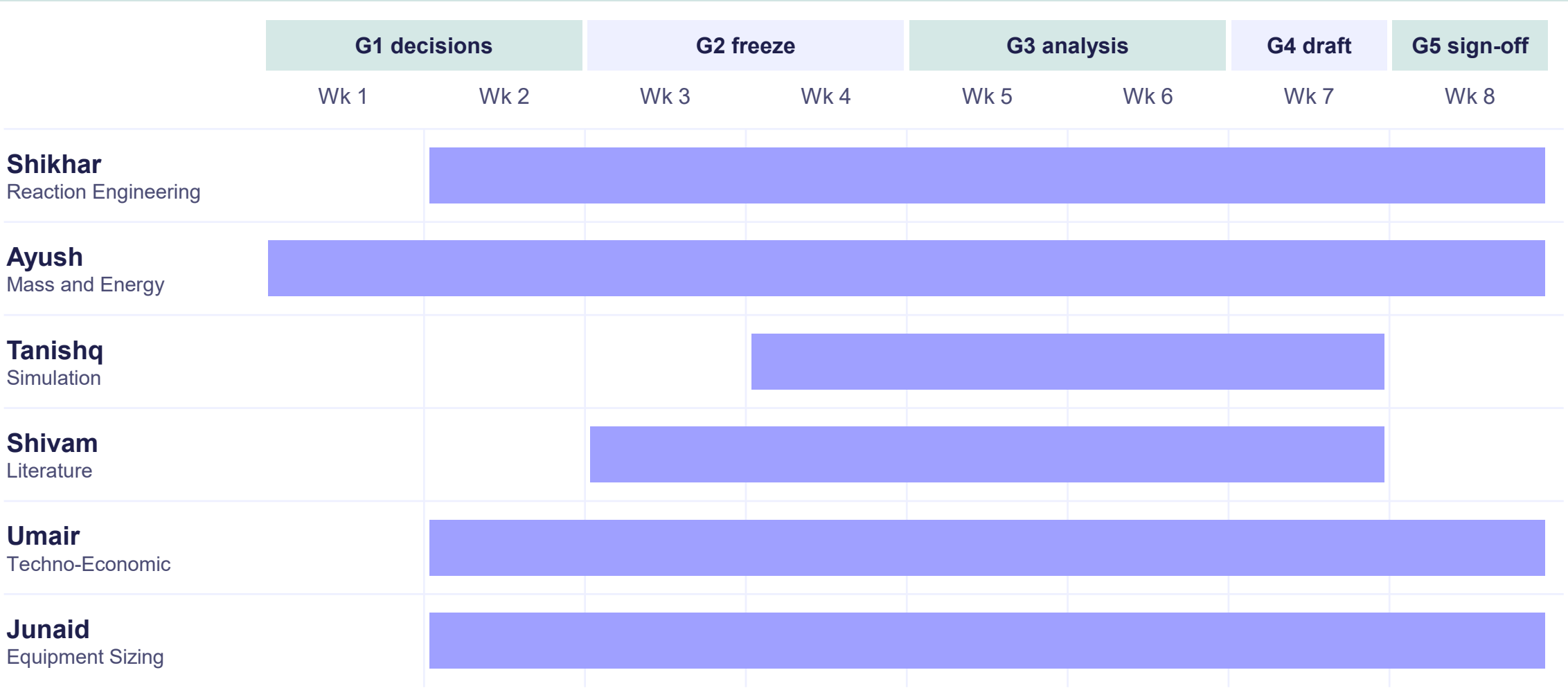
Stage	Outlet (bar)	Outlet (°C)	Power (kW)
1	0.034	74	2.18
2	0.180	189	2.98
3	0.949	189	2.98
4	5.000	189	2.96
Total			**11.10**

Limits of this first pass

U = 250 W/m²K and LMTD = 30 K are defaults, not a thermal design. The electrolyser, gas separator and storage tanks are not sized yet, and they will drive the hardware mass.

Gantt chart

Project timeline and review gates



References

Sources

1. NASA JPL/MIT: MOXIE, Perseverance rover results (2021–2023)
2. Andersen et al. (2024): NASA JSC ISRU trade study, NTRS 20240014064, ASCEND 2024
3. NASA NTRS 20170005179: ISRU propellant production system for a Mars Ascent Vehicle
4. Polsgrove, T. et al. (2017): AIAA 2017-0423
5. NASA NTRS 20160006324: Sustaining human presence on Mars using ISRU and a reusable lander
6. SpaceX: Raptor and Starship documentation
7. ESA: Space Resources Strategy (2019)
8. Guerra, L. and Shishko, R. (2000): Advanced Missions Cost Model
9. Koschany, F. et al. (2016): LHHW kinetics for Ni CO_2 methanation; Xu and Froment (1989)
10. Sabatier, P. and Senderens, J.B. (1902): Comptes Rendus Acad. Sci. 82
11. Goodwin, D. G. et al.: Cantera 3.2.0
12. Bell, I. H. et al. (2014): CoolProp, Ind. Eng. Chem. Res.
13. NIST Chemistry WebBook; IUPAC standard atomic weights
14. Tsiolkovsky, K.: rocket equation and interplanetary mass budgets
15. DWSIM open-source process simulator: documentation, Peng-Robinson package
16. Hu, J. et al. (2007); Strucks, P. et al. (2021): Ni methanation kinetics reviews

Plant simulator website

https://chemical-technology-project.vercel.app

GitHub repository

https://github.com/oyetanishq/chemical-technology-project